Modernizing Retention

Applying High-Frequency Subscription Logic to Personal Lines Insurance

Demonstrating SQL, Statistical Analysis, and Financial Impact Modeling

Project Overview

This project demonstrates a cross-industry retention analytics framework by building a telecom customer churn prediction model and translating its insights into actionable strategies for the auto insurance industry, specifically Progressive Insurance.

Business Problem

- Churn = forfeited years of recurring revenue in telecom
- In auto insurance: policyholder attrition impacts premium volume, loss ratios & long-term profitability
- Understanding *why* customers leave—and *when* to intervene—drives scalable retention

Approach

225,000-row synthetic telecom dataset

1. **EDA — Churn Drivers**
   Contract type, pricing changes, early service friction
2. **XGBoost Model**
   ROC AUC 0.84 · Top 10% = 1.84× baseline churn rate
3. **SHAP Explanations**
   Transparent, interpretable feature-level predictions
4. **Financial Quantification**
   "Friction Churners" = **$32.45M** CLTV at risk (24-mo)

✓ **Save Rate Scenario**

20% save rate → **$6.49M** retained revenue

Cross-Industry Translation

Telecom		Auto Insurance
Promo-to-full-rate	→	Intro rate to renewal price shock
Unresolved billing/tech calls	→	Early claims or underwriting delays
Month-to-month contracts	→	6-month vs annual renewals
Sales channel mix	→	Direct vs agent vs aggregator

Key Findings & Strategic Value for Progressive

Key Findings

72.8%
month-to-month

Contract Type Drives Churn

Month-to-month customers churn at **72.8%** vs. only **24%** for annual/two-year contracts — a **3× difference** in retention risk.

3×
higher risk

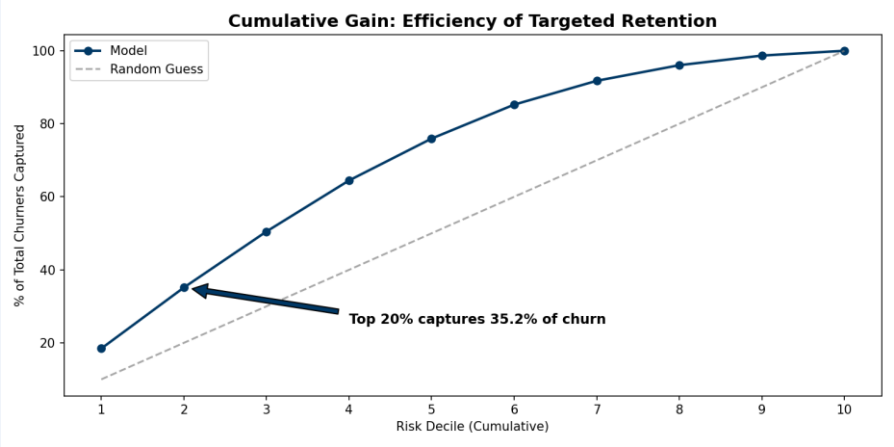
16.7%
preventable

Early Friction Churn (First 60 Days)

16.7% of all churn stems from unresolved service issues in the first 60 days — representing **preventable loss** addressable through rapid resolution protocols.

Top 20% Captures 35.2% of Churn

XGBoost model (AUC 0.84) enables precise targeting — the highest-risk quintile captures **35.2% of all churners** at 1.84× baseline lift.



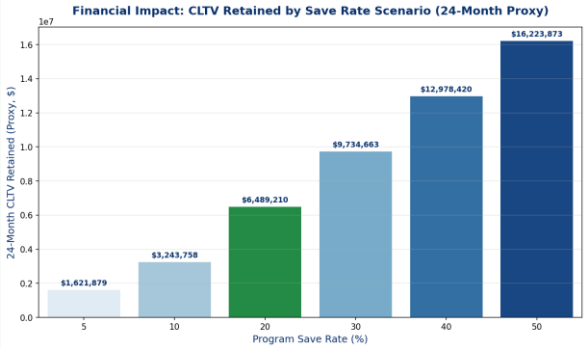
Strategic Value for Progressive

1 Identify High-Risk Policyholders

Predictive model identifies top 20% highest-risk segment with **1.84× baseline churn rate**, enabling proactive intervention **30 days before renewal**.

2 Quantify Financial Impact

$32.45M in 24-month CLTV at risk. A 20% save rate retains **$6.49M** in customer lifetime value.



3 Prioritize Operational Investments

Rapid Resolution team targeting first 60-day friction (**16.7% of churn**) with 48-hour SLA. Contract incentives reduce churn from **72.8%** to **24%**.

4 Cross-Industry Analytics Transfer

Telecom churn drivers — **contract type, pricing changes, early friction** — map directly to insurance levers: **policy terms, rate disruption, claims experience**.

Telecom Churn Prediction – Cross-Industry Retention Framework

- XGBoost model trained on 225,000 synthetic customers · Portfolio case study targeting Progressive Insurance
- Synthetic dataset · SHAP values computed on held-out test set · Financial model assumes $1,695 avg. CLTV per friction churner

Model Performance

Algorithm	**XGBoost**
Training Set	**225K**
ROC AUC	**0.84**
Top Decile Lift	**1.84×**
Top 20% Captures	**35.2%**

Top Churn Drivers · SHAP

1. **Contract Type** #1 Factor
   Month-to-month: 72.8% churn vs 24% annual
2. **Monthly Charges**
   Higher bills → higher flight risk
3. **Early Friction**
   16.7% of churners leave in first 60 days
4. **Tech Support Absence**
   No support → elevated churn
5. **Sales Channel**
   Direct vs agent vs aggregator mix

Financial Impact

Value Retained
$6.49M ✓

Friction Churners	**19,147**
CLTV at Risk	**$32.45M**
Save Rate	**20%**
Customers Retained	**3,829**

Cross-Industry Retention Playbook

Every telecom churn signal has a direct parallel in auto insurance retention

	Telecom Churn Driver		Progressive Insurance Analogue
1	Month-to-month contracts → 72.8% churn rate	→	6-month policy terms vs annual renewals → shorter commitment window
2	Promotional pricing → full rate shock	→	Intro rate → renewal price shock for new policyholders
3	Early billing / tech friction (first 60 days)	→	First 60-day claims experience issues → early attrition signal
4	Sales channel mix (direct / agent / aggregator)	→	Direct vs independent agent vs online aggregator acquisition
5	Autopay & paperless enrollment	→	EFT, automatic renewal, e-delivery → engagement & stickiness
6	Streaming / add-on service bundles	→	Multi-product bundling (auto + home + renters)
7	Fiber optic market competition	→	Competitive metro markets → price-sensitive churn pressure

This framework is portable — the same SHAP-ranked feature logic deployed on telecom data can be retrained on Progressive's policy, claims, and billing data to produce a retention score for every policyholder at renewal.

Three Retention Levers to Deploy

- Data-driven recommendations built directly from model outputs and SHAP analysis
- All projections derived from XGBoost model outputs · Save rate assumption: 20% · CLTV basis: $1,695 per friction churner

01

Early Friction Resolution Program

- Identify customers with billing or service friction in first 60 days
- Deploy proactive outreach within 48-hour SLA
- Target: 19,147 friction churners in telecom model

Projected Value Retained

$6.49M

@ 20% save rate · 3,829 customers

02

Contract Term Incentives

- Offer discount for 12-month policy + autopay enrollment
- Address root cause: 72.8% churn on short-term vs 24% on annual
- Parallels telecom month-to-month → annual contract migration

Churn Rate Reduction Target

72.8% → ~24%

Short-term to annual commitment

03

Predictive Targeting Model

- Score every policyholder at renewal using trained XGBoost model
- Focus outreach on top 20% risk segment
- Launch proactive retention 30 days before renewal date

Churner Capture Rate

35.2%

Of all churners in top 20% risk tier

Thank You

I appreciate your time and consideration

This mock project demonstrates my approach to predictive modeling and financial impact analysis: rigorous statistical methodology, comprehensive data storytelling and actionable business insights.

Let's Connect

Email: luciano.casillasjr@outlok.com

Phone: 512-534-1254

LinkedIn: linkedin.com/in/luciano-casillas-65675310

I look forward to discussing how I can contribute to your team.

Appendix

Analytics Roadmap

Six-phase end-to-end methodology · Telecom Churn Prediction project

01

Synthetic Data Generation

225,000 rows · Realistic telecom customer profiles with behavioral signals

Foundation

02

Exploratory Data Analysis

5 key figures · Churn distributions, feature correlations, segment analysis

Insight

03

Feature Engineering

Interaction terms, early friction flag, contract risk score, channel encoding

Modeling Prep

04

XGBoost + SHAP Analysis

ROC AUC 0.84 · Top Decile Lift 1.84× · SHAP feature importance ranking

Core Model

05

Financial Impact Simulation

$32.45M CLTV at risk · $6.49M value retained at 20% save rate

Business Value

06

Executive Summary & Translation

Progressive Insurance analogue mapping · Three actionable retention levers

Delivery

Full reproducible code available · Python · XGBoost · SHAP · Pandas · Matplotlib · Portfolio project